MONEY OR EDUCATION: WHAT DRIVES MATHS RESULTS?

PISA 2025 · sociology group project

Presented By
OUR TEAM

NEXT

HYPOTHESIS AND MAIN FINDING

Hypothesis: parental wealth affects children's results more than parental education.

01 **Parental education is the stronger predictor in 59 of 87 countries.**

02 **Wealth is stronger in 28 countries, mostly low-resource ones.**

03 **Together, the two factors explain only about 10% of score differences. The other 90% depends on schools, teachers, motivation, ability and chance.**

INTRODUCTION



On an interactive map, you can specify the living conditions of a 15-year-old student and see the average PISA 2025 mathematics score achieved by students with those conditions in each country.

Ukraine (selected regions, QUA)

country average: 431

Predicted maths score

435 points

−3	+82	+39
from the poorest 25% to the richest 25% (same education and culture)	from basic to higher parental education (same money and culture)	from the lowest to the highest cultural capital (everything else the same)

In this country **parental education matters more** than money.

Prediction for all combinations (cultural capital: нижче середнього). Rows are economic capital, columns are parental education. Click a cell to select it.

	базова	середня	вища
найбагатші 25%	363	422	446
вище середнього	372	430	454
нижче середнього	376	435	459
найбідніші 25%	367	425	449

CHILD'S CONDITIONS

◂ Hide

Parental education — secondary

Economic capital — below average

number of cars, rooms with a bath or shower, and rooms with a toilet at home

Cultural capital — below average

number of books, musical instruments and works of art at home

☑ Include cultural capital. If switched off, the model does not know about it, and part of its effect "moves" to education and economic capital.

WHAT THE MAP SHOWS

Prediction | Average score

COUNTRY

Ukraine (selected regions, QUA)

Or click a country on the map. Small countries (Singapore, Macao, Malta…) are easier to pick from the list.

DATA AND METHOD

PISA 2025 file: 747,000 students in 89 countries and economies. The analysis uses 636,000 students (85%) from 87 countries with known parental education and wealth.

01

Indicators: maths score (10 plausible values per student), parental education in years of schooling (6, 9, 12, 14, 16), the home possessions index HOMEPOS, the composite ESCS index, and the number of books at home.

02

All calculations run separately for each country, with student weights and for each of the 10 plausible values, then averaged. Countries count equally.

03

Strength of a factor: standardized regression coefficients and the share of score differences explained (R^2).

04

OVERALL PICTURE

- In raw terms, the two gaps are almost identical: the poorest 20% of homes trail the richest 20% by about 54 points, and children of parents with up to 9 years of schooling trail those with 16 years by about 54 points. That is roughly two to three school years of learning.
- Median scores rise about 55 points from the bottom group to the top on both measures.
- Share of students below about 420 points (roughly the basic proficiency threshold): 60% in the poorest fifth and 37% in the richest; 74% for children of parents with up to 9 years of schooling and 34% for those with 16 years.
- The groups overlap heavily: family background shifts the distribution without deciding an individual child’s result.

EDUCATION VS WEALTH IN ONE MODEL

Both factors together explain **9.7%** of score differences.

Average across countries	Parental education	Home wealth (HOMEPOS)
Standardized coefficient	0.20	0.15
Maths points per 1 standard deviation	about 18	about 12.5
Share of differences explained, alone	7.2%	5.7%
Countries where the factor is stronger	59 of 87	28 of 87

COUNTRY DIFFERENCES

Low-resource third of countries: wealth is stronger in 20 of 29 (0.19 against 0.15 for education).
Middle and upper thirds: education leads (0.23 against 0.13, and 0.23 against 0.12); wealth is stronger in only 8 of 58 countries, usually by a small margin (Italy, Austria, New Zealand, the USA).
Largest wealth advantages: Zambia, Malaysia, Colombia, Ecuador, Brazil (also Vietnam, Argentina, Peru, Morocco).
Education dominates most strongly: the UAE, Qatar, Macao, Canada, Taipei, Malta.
Practical reading: where material conditions at home are uncertain, money and possessions become the main barrier. Where basic conditions are secured for almost everyone, what decides the result is whether the family can help with learning.

WHAT “WEALTH” REALLY MEASURES

HOMEPOS MIXES MATERIAL POSSESSIONS WITH BOOKS. SPLIT INTO PARTS (82 COUNTRIES):

books at home: **0.195**

parental education: 0.162

material possessions: 0.110

Books are the strongest single factor in 47 countries, parental education in 20, and material possessions in 15. So the “effect of wealth” is to a large extent, the effect of a home learning environment, which correlates with money but is not the same thing.

Correlations with maths score: ESCS 0.32; books 0.28; parental education 0.26; HOMEPOS 0.22; material items only 0.22. Parental education and wealth correlate at 0.34, so they carry largely different information. The spread across countries is wide, so the ranking should not be read as exact in any single country.

Higher education partly compensates for poverty. Children of university-educated parents in the poorest fifth trail less than children of parents with up to 9 years of schooling at middle wealth.

Money without education helps little. Among parents with up to 9 years of schooling, the poor–rich gap is only 21 points; among university-educated parents it is 51 points.

Atypical groups are small: poorly educated but richest families are 1.2% of students, educated but poorest families 5.1%. The largest group is educated and well-off families, at 13.1%.

In low-resource countries high parental education barely rescues a poor family: the poorest-fifth children of graduates trail those of parents with up to 9 years of schooling by only about 6 points, against about 53 in high-resource countries.

EDUCATION AND WEALTH TOGETHER

The factors reinforce each other: the best results belong to children of educated and well-off parents, the worst to children of poorly educated and poor ones, a gap of 79 points.

LIMITATIONS

- Correlations, not causes. The data do not show that raising parents' income or education would raise children's scores. Ability and motivation may be passed on within families.
- Wealth is measured weakly. HOMEPOS counts household items reported by the 15-year-old student, not income or assets, and it includes books. Parental education has only five levels. Even so, on equal terms (9 education categories against 9 wealth groups) education explains more: 11.2% against 7.4%, and in 72 of 87 countries.
- Missing data are not random. Parental education is missing for 14.6% of students, who score on average 40 points below their country's average. The most disadvantaged children are underrepresented, so real gaps are probably larger.
- Development level is approximated by average HOMEPOS, not GDP. The data were not checked against official documentation.

IDEAS FOR SOLUTIONS

A. Remove material barriers

- free basic “learning pack”: textbooks, notebooks, a device, internet
- free school meals
- free clubs and extended-day programmes
- needs-based school funding

B. Replace the missing home learning capital

- small-group tutoring (2–4 students) for those who fall behind
- books at home and school libraries, paired with reading activities
- free or subsidized preschool for children from poor families
- diagnostic tests at the end of primary school and automatic extra support

C. Work with parents

- programmes for parents with low education
- adult education
- transparent, plain-language school communication

D. System-level decisions

- less school segregation
- the best teachers in the hardest schools
- scholarships and mentoring for capable children from poor families

CONCLUSION

9 OUT OF 87 COUNTRIES WHERE WEALTH MATTERS MORE THAN PARENTS' EDUCATION

78 OUT OF 87 COUNTRIES WHERE PARENTS' EDUCATION MATTERS MORE THAN WEALTH

77 OUT OF 87 COUNTRIES WHERE CULTURAL CAPITAL MATTERS MORE THAN WEALTH

We compared children with the same parental education and cultural capital. Moving from the poorest 25% of families to the wealthiest 25% results in a greater increase in scores than moving from basic to higher parental education in only 9 out of 87 countries. If cultural capital is not taken into account, this is the case in 11 countries.

This means that in most countries, the hypothesis is not supported: parental education and family cultural capital are more strongly associated with scores than material possessions and money. However, there are countries where wealth matters more than education, such as Lebanon, Cambodia, Zambia, Argentina, Ecuador, and Malaysia. These are mostly lower-income countries, where a family's wealth can determine whether a child has access to a quality school at all.

THANK YOU

FOR YOUR ATTENTION!